Ablation performance

Six variants evaluated on the selected graph.

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.879 ± 0.003**	0.802 ± 0.003	0.878 ± 0.019	0.853 ± 0.006	0.872 ± 0.002	0.862 ± 0.011
10	0.876 ± 0.003	0.806 ± 0.011	**0.880 ± 0.010**	0.853 ± 0.004	0.872 ± 0.003	0.861 ± 0.009
15	0.873 ± 0.004	0.812 ± 0.002	**0.878 ± 0.009**	0.852 ± 0.004	0.871 ± 0.003	0.860 ± 0.008
20	0.873 ± 0.003	0.818 ± 0.006	**0.877 ± 0.009**	0.852 ± 0.004	0.871 ± 0.003	0.858 ± 0.007
25	0.873 ± 0.003	0.825 ± 0.004	**0.877 ± 0.009**	0.851 ± 0.004	0.871 ± 0.002	0.858 ± 0.006
30	0.872 ± 0.002	0.829 ± 0.002	**0.877 ± 0.008**	0.851 ± 0.003	0.871 ± 0.002	0.857 ± 0.005
35	0.871 ± 0.002	0.832 ± 0.001	**0.877 ± 0.008**	0.851 ± 0.003	0.870 ± 0.002	0.857 ± 0.004
40	0.871 ± 0.002	0.837 ± 0.001	**0.876 ± 0.007**	0.851 ± 0.003	0.870 ± 0.002	0.856 ± 0.004
45	0.870 ± 0.003	0.842 ± 0.000	**0.876 ± 0.006**	0.851 ± 0.003	0.869 ± 0.002	0.856 ± 0.004
50	0.870 ± 0.003	0.845 ± 0.000	**0.876 ± 0.005**	0.850 ± 0.003	0.869 ± 0.002	0.855 ± 0.003

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**1.000 ± 0.000**	1.000 ± 0.000	0.998 ± 0.001	0.999 ± 0.001	1.000 ± 0.000	1.000 ± 0.000
10	**1.000 ± 0.000**	1.000 ± 0.000	0.997 ± 0.001	0.999 ± 0.001	1.000 ± 0.000	1.000 ± 0.000
15	1.000 ± 0.000	**1.000 ± 0.000**	0.997 ± 0.002	0.999 ± 0.001	1.000 ± 0.000	1.000 ± 0.000
20	1.000 ± 0.000	**1.000 ± 0.000**	0.996 ± 0.002	0.999 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
25	1.000 ± 0.000	**1.000 ± 0.000**	0.996 ± 0.002	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
30	1.000 ± 0.000	**1.000 ± 0.000**	0.994 ± 0.002	0.999 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
35	1.000 ± 0.000	**1.000 ± 0.000**	0.994 ± 0.001	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
40	1.000 ± 0.000	**1.000 ± 0.000**	0.993 ± 0.002	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
45	1.000 ± 0.000	**1.000 ± 0.000**	0.992 ± 0.003	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
50	1.000 ± 0.000	1.000 ± 0.000	0.992 ± 0.003	1.000 ± 0.000	1.000 ± 0.000	**1.000 ± 0.000**

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Comparison performance

Eight baselines and 2CKG4ER evaluated on the selected graph.

(a) Adaptability (Ada)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.821	0.834	0.549	0.844 ± 0.003	0.863 ± 0.004	0.789 ± 0.010	0.862 ± 0.003	0.856 ± 0.004	**0.879 ± 0.003**
10	0.811	0.840	0.577	0.849 ± 0.002	0.862 ± 0.002	0.816 ± 0.006	0.861 ± 0.002	0.855 ± 0.003	**0.876 ± 0.003**
15	0.811	0.844	0.594	0.850 ± 0.001	0.860 ± 0.001	0.827 ± 0.006	0.859 ± 0.002	0.854 ± 0.001	**0.873 ± 0.004**
20	0.813	0.846	0.604	0.850 ± 0.001	0.858 ± 0.003	0.833 ± 0.006	0.857 ± 0.002	0.852 ± 0.001	**0.873 ± 0.003**
25	0.815	0.846	0.613	0.848 ± 0.002	0.855 ± 0.002	0.836 ± 0.006	0.857 ± 0.002	0.850 ± 0.000	**0.873 ± 0.003**
30	0.817	0.847	0.621	0.846 ± 0.002	0.853 ± 0.001	0.837 ± 0.005	0.855 ± 0.003	0.848 ± 0.001	**0.872 ± 0.002**
35	0.818	0.845	0.628	0.845 ± 0.002	0.852 ± 0.001	0.838 ± 0.005	0.855 ± 0.003	0.847 ± 0.002	**0.871 ± 0.002**
40	0.820	0.842	0.633	0.846 ± 0.001	0.851 ± 0.001	0.839 ± 0.005	0.854 ± 0.004	0.846 ± 0.001	**0.871 ± 0.002**
45	0.821	0.840	0.638	0.845 ± 0.001	0.851 ± 0.001	0.839 ± 0.004	0.854 ± 0.004	0.846 ± 0.002	**0.870 ± 0.003**
50	0.822	0.839	0.642	0.846 ± 0.001	0.851 ± 0.001	0.838 ± 0.004	0.853 ± 0.004	0.845 ± 0.002	**0.870 ± 0.003**

(b) Novelty (NOV)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.998	1.000	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
10	0.998	1.000	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000
15	0.999	1.000	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000
20	0.999	0.999	1.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	1.000 ± 0.000	1.000 ± 0.000
25	0.998	0.999	1.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	1.000 ± 0.000	1.000 ± 0.000
30	0.996	0.998	1.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	1.000 ± 0.000	1.000 ± 0.000
35	0.994	0.997	1.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	1.000 ± 0.000	1.000 ± 0.000
40	0.993	0.998	1.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	**1.000 ± 0.000**	1.000 ± 0.000	1.000 ± 0.000
45	0.993	0.998	1.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
50	0.992	0.998	1.000	**1.000 ± 0.000**	**1.000 ± 0.000**	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.